

Camera Sensor Intrinsic Calibration Uploader User Guide

ExcelSense Technologies Corp.

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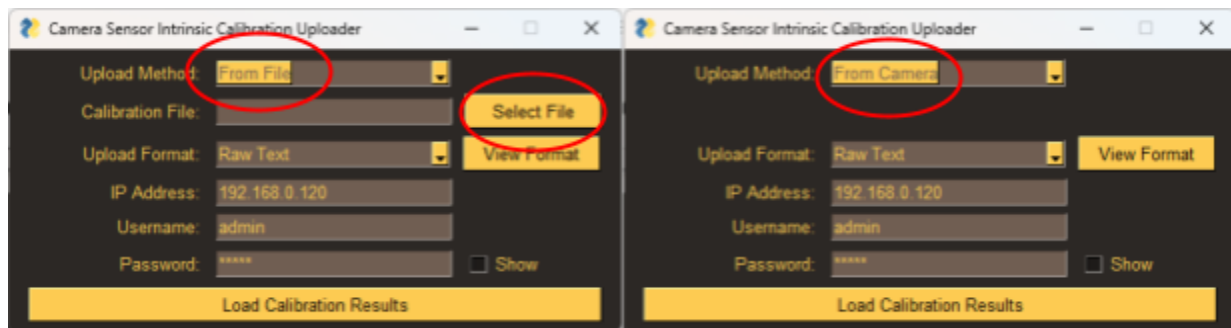
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Background

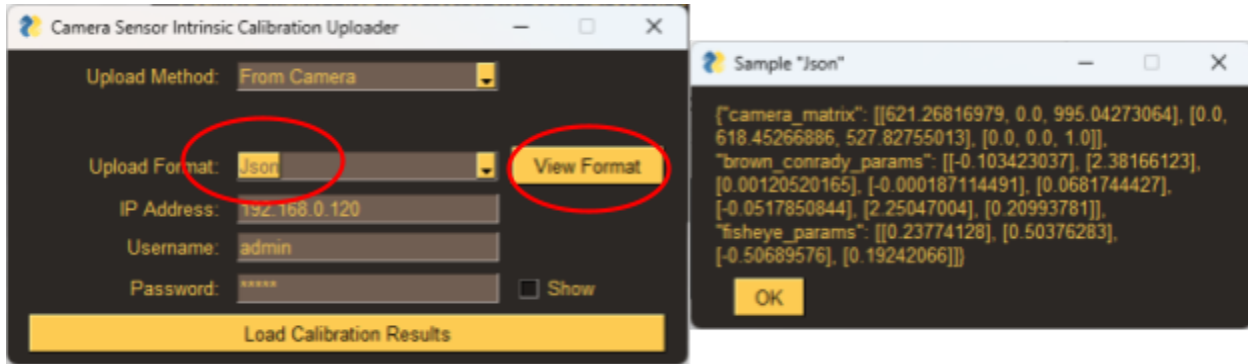
This document outlines the procedure for uploading the intrinsic camera sensor calibration from a .txt file or replacing the resulting format found inside the camera using the Camera Sensor Intrinsic Calibration Uploader Application. More information about intrinsic calibration procedures can be found in the *Camera Sensor Intrinsic Calibration Procedure* documentation.

Upload Procedure

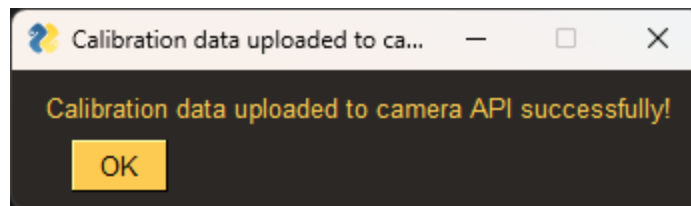
The user can upload the calibration results by loading them from a file, or by reading the results that were previously saved in the camera.



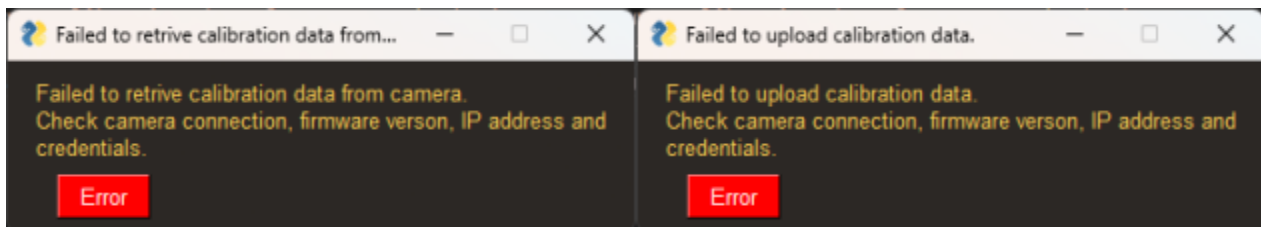
The user can also see samples of available calibration data formats to help decide which works best for their application by selecting an upload format and clicking the button *View Format*. The available data formats are shown in the next Section.



Next, the user selects the Camera's IP address and types the username and password and clicks the button *Load Calibration Results*. If no error is encountered, then the calibrated data will be correctly uploaded to the camera's internal memory.



If the app cannot read the file format stored in the camera, or if it fails to upload data to the camera, the following errors may occur. In such cases, the user should ensure that the camera connection is stable and that the file format in the camera is supported by the app.



Sample Data

Below are the sample data formats the user can choose from when uploading the calibration results or converting the saved data inside the camera to a different format. This data string can be retrieved through the existing network connection by making an HTTP request to the camera's API, as described in the *Camera Sensor Intrinsic Calibration Procedure* documentation.

Raw Text

This is the raw text found in the calibration results text file.

```
Calibration Results:

Calibration Time: 288.34768748283386

Camera Matrix:
[[621.26816979    0.          995.04273064]
 [   0.          618.45266886  527.82755013]
 [   0.           0.           1.          ]]

Brown-Conrady Distortion Parameters:
[[-1.03423037e-01]
 [ 2.38166123e+00]
 [ 1.20520165e-03]
 [-1.87114491e-04]
 [ 6.81744427e-02]
 [-5.17850844e-02]
 [ 2.25047004e+00]
 [ 2.09937810e-01]]

Fisheye Distortion Parameters:
[[ 0.23774128]
 [ 0.50376283]
 [-0.50689576]
 [ 0.19242066]]
```

Pretty Text

This is a prettier version than the raw text data, which compacts the calibration results and removes unnecessary text.

```
Camera Matrix:
[[621.26816979    0.          995.04273064]
 [   0.          618.45266886  527.82755013]
 [   0.           0.           1.         ]]

Brown-Conrady Distortion Parameters:
[[-1.03423037e-01]
 [ 2.38166123e+00]
 [ 1.20520165e-03]
 [-1.87114491e-04]
 [ 6.81744427e-02]
 [-5.17850844e-02]
 [ 2.25047004e+00]
 [ 2.09937810e-01]]

Fisheye Distortion Parameters:
[[ 0.23774128]
 [ 0.50376283]
 [-0.50689576]
 [ 0.19242066]]
```

Json String

Typical json string with named variables for ease of use and parsing with python libraries.

```
{"camera_matrix": [[621.26816979, 0.0, 995.04273064], [0.0, 618.45266886, 527.82755013], [0.0, 0.0, 1.0]], "brown_conrady_params": [[-0.103423037], [2.38166123], [0.00120520165], [-0.000187114491], [0.0681744427], [-0.0517850844], [2.25047004], [0.20993781]], "fisheye_params": [[0.23774128], [0.50376283], [-0.50689576], [0.19242066]]}
```

Custom Json String Separated by “|”

Custom json string with minimal information for reconstructing the calibration matrix and other parameters. Each array is separated by the character “|”.

```
{"fx":621.26816979,"fy":618.45266886,"cx":995.04273064,"cy":527.82755013}|  
{"k1":0.23774128,"k2":0.50376283,"k3":-0.50689576,"k4":0.19242066}|{"k1":-  
0.103423037,"k2":2.38166123,"p1":0.00120520165,"p2":-0.000187114491,"k3":0  
.0681744427,"k4":-0.0517850844,"k5":2.25047004,"k6":0.20993781}
```